



Program: B.Tech in Computer Science and Engineering (DataScience)

Semester: IV

Course: Web Engineering Laboratory (DJS23DMD251L)

Pre-requisite: Programming Fundamentals

Objectives:

To provide the basic framework of web development (MERN Stack) and cloud computing.

Outcomes: On completion of the course, the learner will be able to:

1. Develop a website as per the requirements.
2. Apply the concepts of cloud computing to improve the efficiency of web development.
3. Evaluate the requirement of the problem and select appropriate method of web development.

Web Engineering Laboratory (DJS23DMD251L)		
Unit	Description	Duration
1	HTML: 1. Create a static web page using HTML. 2. Create a class timetable using HTML. 3. Create a registration form using HTML. 4. Create a web page using HTML5 tags.	02
2	CSS: 1. Design a responsive web page and CSS3 using external CSS (media queries). 2. Design a web page using Bootstrap. 3. Design the admission form using Bootstrap.	02
3	Client-Side Scripting: 1. Programs based on objects in JavaScript. 2. Programs based on form validation. 3. Fetch Data from an API Using the Fetch API in JavaScript.	06
4	React JS: 1. Types of Components 2. Single Page Application (Virtual DOM) 3. State 4. React Hooks	08
5	Introduction to Git and GitHub: 1. Adding files to the staging area 2. Push new branch 3. Merge into the master 4. Auto Commit	02
6	Server-Side Scripting: 1. Installation and Configuration of Node.js server 2. Program based on inbuilt functions in Node.js	04
7	Express & MongoDB: 1. Using Mongoose to make schemas in MongoDB. 2. Making API end points using Express and testing using postman. 3. Develop a website and integrate it with pre-defined API.	08



	4. Doing CRUD on database MongoDB using Express. 5. PostgreSQL & Sequelize	
8	XML & XSL: - Design XML using XML DTD and schema. - Implementing XSL elements in XML. - Validate XML using Node.js	04
9	Concepts of Cloud Computing: - Introduction to cloud computing. - NIST model - Service and Deployment models.	04
10	Networking and Security: - Identity and Access Management - Networking basics - VPC networking and security - Design a VPC - Build your own VPC and Launch a Web Server	04
11	Compute Service: - Compute Services overview - Elastic Computing - Serverless Compute service - Deploying and scaling web applications	04
12	Storage Service: - Cloud object storage - Cloud block storage - Elastic file system	04
13	Database Service: - Cloud Relational database services - Cloud NoSQL Databases - Elastic load balancing	04
	Total	56

*The Term Work will be calculated based on Laboratory Performance (15m), Mini Project (10m) and Computer Based Assessment (25m).

Books Recommended:

Text books:

- Vasan Subramanian, Pro MERN Stack, 2nd Edition, Apress Publication, 2019.
- Shama Hoque, Full-Stack React Projects, 2nd Edition, Packt Publication, 2020
- Rajkumar Buyya, James Broberg, Goscinski Cloud Computing: Principles and Paradigms, Wiley, 2013.
- Shalabh Aggarwal, Flask Framework Cookbook: Over 80 proven recipes and techniques for Python web development with Flask, Packt publication, 2nd Edition, 2019.

Reference Books:

- Benjamin LaGrone, HTML5 and CSS3 Responsive Web Design Cookbook, 1st Edition, Packt Publishing, 2013.



2. DT Editorial Services, Web Technologies: Black Book, 1st Edition, Dreamtech Press, 2018.
3. Christopher Schmitt, Kyle Simpson, HTML5 Cookbook, 1st Edition, O'Reilly Media Inc., 2011.
4. Uttam K. Roy, Web Technologies, 1st Edition, Oxford University Press, 2010.
5. Greg Sidelnikov, React. Js Book: Learning React JavaScript Library from Scratch, 1st Edition, Independently Published, 2017.
6. DT Editorial Services, HTML5 Black Book, 2nd Edition, Dreamtech Press, 2016.
7. Ben Frain, Responsive Web Design with HTML5 and CSS3, 2nd Edition, Packt Publishing, 2015.
8. Steve Suehring, JavaScript Step by Step, 3rd Edition, Pearson Education, 2013.
9. Stoyan Stefanov, React Up Running Building Web Applications, 1st Edition, O'Reilly Media Inc., 2016.
10. Velte, Cloud Computing a Practical Approach, Tata McGraw-Hill Education.
11. Sandip Bhowmik, Cloud Computing, Cambridge University Press, 2017.
12. Miguel Grinberg, Flask Web Development, O'Reilly publication, 2018
13. Sack Stouffer Daniel Gaspar, Mastering Flask Web Development, Packt Publication, 2018

Web Links:

1. Professional and Lifelong Learning: [Web Development Courses | Harvard University](#)
2. Virtual Lab: <https://html-iitd.vlabs.ac.in/exp/introduction-to-html/references.html>